# Module_Recon Quality-Of-Life Suggestions

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Companion to `Review.md` and `Review-Clarifications.md`. Where the clarification register asks "how should the move work", this document asks "what else should the module catch". Everything here is optional. Nothing here is a code change request by itself.

## Framing

The current feature answers one question: *"did the saved location drift from where Infor Visual actually put the material?"*

Most real receiving exceptions are not location drift. They are:

- a label that was never printed,
- a label printed twice,
- two labels for the same physical skid,
- a part received without a label at all,
- a quantity entered differently from what the ERP recorded,
- a heat lot that should exist and does not,
- a row saved against a closed or wrong PO line.

A reconciliation module that only fixes locations leaves those on the floor. The suggestion below is to grow the module from a *location repair tool* into a *receiving integrity auditor* whose output is a prioritized exception list, of which "wrong location" is one exception type among several.

## Design guardrail

Every detector below must follow the same contract as location reconciliation:

1. **Detect** – produce evidence-backed findings.
2. **Never silently write** – anything that changes data goes through an explicit user action.
3. **Fail visible** – unresolved findings are surfaced, never swallowed.
4. **Stay read-only against Infor Visual.**

5. **Explain itself** – every finding carries a human-readable reason and its evidence.

Reusing the existing finding model (existing location, proposed location, resolution, details, evidence) means most detectors can reuse the same grid, filters, and export.

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# 1. A. Label Presence And Print Integrity

These are the "did the paperwork match the material" checks. None of them exist today.

## A1 — Missing label (received but never labeled)

**Catches:** material was received in Infor Visual but the operator never created a label row, or created fewer labels than the receipt requires.

**Method:** compare ERP receipt quantity against the sum of MTM label quantities for the same `(PO, part, line, receipt window)`:

```
erp_received_qty  vs  sum(mtm_label.weight_quantity)  for the same PO + part + line
```

Outcome buckets: `FullyLabeled`, `UnderLabeled` (short by N), `NotLabeled` (no rows at all), `OverLabeled` (see A3).

**Data needed:** ERP receipt quantity ( `RECEIVER_LINE` / `PURC_ORDER_LINE` received quantity), or, without ERP access, `qty_ordered` minus `remaining_quantity` as a weaker proxy. The proxy is weaker because partial receipts and multiple receipts against one line are indistinguishable.

**Effort:** M (query work). **Value:** high — this is the failure mode that produces unlabeled floor stock. **Risk:** medium — false positives if ERP quantity is not scoped to the same receipt.

**Depends on:** `LG-02` (receiver data source). If the answer is "MTM rows are the receiver proxy", this detector cannot be built and must be dropped.

## A2 — Label never printed

**Catches:** a label row exists in `receiving_label_data` but no label was ever sent to the printer, so a skid sits on the floor with no tag.

**Blocker:** `receiving_label_data` has **no** print timestamp and no print audit table. This cannot be inferred reliably today.

**Options:**

- **A2a** Add `printed_at DATETIME NULL` and `print_count INT` to `receiving_label_data` plus a write in the print path. Simple, exact going forward, no history.
- **A2b** Add a separate `receiving_label_print_log` table (row id, printed_at, employee, printer, is_reprint). Preserves history and supports A5/A6.
- **A2c** Infer from the label-overlay print service call only at print time (no persistence), which gives a one-session "not yet printed" badge but no audit.

**Recommendation:** A2b. A2a is the cheapest but loses the ability to answer "printed twice?". A2c is a UI nicety, not an integrity check.

**Effort:** M (schema + print path + detector). **Value:** high. **Risk:** low.

## A3 — Duplicate or double-counted labels

**Catches:** the same physical skid represented by more than one row, so quantity is overstated.

Signals already available:

- `is_reprint = 1` rows sitting in `receiving_label_data` alongside the original history row.
- Identical `(part_id, po_number, po_line_number, weight_quantity, initial_location)` rows created
  within a short time window.

- `load_id` duplicated across rows that should be distinct skids.

**Method:** flag groups where the combined quantity exceeds the transfer or receipt evidence, and flag explicit reprint rows that share an identity with a history row.

**Effort:** S. **Value:** high — this is a silent quantity-inflation bug class.
**Risk:** low, but the reason text must distinguish "possible reprint duplicate" from "possible double entry" or operators will distrust it.

**Note:** this detector is what makes `LG-11` (reprint rows in reconciliation) safe either way.

## A4 — Label split integrity

**Catches:** the "N of M" label arithmetic is wrong.

Available columns: `label_number`, `part_skid_sequence`, `part_skid_total`, `packages_per_load`, `weight_per_package`, `weight_quantity`.

Checks:

- `part_skid_sequence` values are `1..part_skid_total` with no gaps or duplicates.
- `sum(weight_quantity)` for a save batch equals the batch total recorded at save time.
- `weight_per_package * packages_per_load` equals `weight_quantity` (within the rounding epsilon).
- `label_number` restarts correctly per load.

**Effort:** S. **Value:** medium (wrong labels get reprinted less often).
**Risk:** low.

## A5 — Reprint abuse / unplanned reprint volume

**Catches:** the same skid reprinted repeatedly, or reprints created outside the Reprint module.

**Data needed:** the print log from A2. Reports reprint count per row, per employee, per day.

**Effort:** S once A2b exists. **Value:** medium.

## A6 — Quality hold not acknowledged

**Catches:** `is_quality_hold_required = 1` with `is_quality_hold_acknowledged = 0` on a row that was already archived to history.

**Effort:** S. **Value:** medium-high (compliance). **Risk:** low.

# 2. B. Cross-Check Against Infor Visual Receipts

These checks compare the MTM row to ERP truth. All are read-only Visual queries.

## B1 — PO line does not exist, or belongs to another part

`Review.md` §7.6 already requires matching on PO + part. This extends it: check that the part is actually on the PO line recorded on the row ( `PURC_ORDER_LINE` ).

**Catches:** typos in PO or line, part substituted at the dock, blanket-order suffix confusion (the `PO-061555B` case in `docs/prompts/reconciliation-location-fix.md` Fix 1).

**Effort:** S (one query). **Value:** high — cheap, and catches data-entry errors that currently survive forever.

## B2 — PO line closed, canceled, or fully received

**Catches:** labels created against a PO line that was already closed, which usually means the receipt was mis-keyed or duplicated.

**Effort:** S. **Value:** medium.

## B3 — Over-receipt against the PO line

**Catches:** total labeled quantity exceeds `qty_ordered - already_received` for the line.

**Effort:** M (needs correct ERP received-quantity semantics). **Value:** high (over-receipt is a payables problem, not just a labeling problem). **Risk:** medium (false positives without partial receipt handling).

## B4 — Received date outside the PO due-date window

**Catches:** labels dated far outside the PO's expected window, which usually means a wrong PO was selected.

**Effort:** S. **Value:** medium. **Risk:** low if the threshold is generous and configurable.

## B5 — Unit-of-measure mismatch

**Catches:** MTM row says `LB` while the Visual part is `EA` (or vice versa).

**Effort:** S. **Value:** high — a UOM mismatch silently invalidates every quantity comparison, including location reconciliation ( `LG-06` ).

## B6 — Part not found in Infor Visual

**Catches:** non-PO or obsolete part numbers that can never be reconciled.

**Effort:** S. **Value:** low-medium. Overlaps the existing `IsNonPOItem` handling.

# 3. C. Row-Level Data Quality

Cheap field checks over rows already loaded. All are `S` effort.

| ID | Check | Catches | Value |
|---|---|---|---|
| C1 | `initial_location` blank | Rows that can never be reconciled or traced | High |
| C2 | `initial_location` not a real Visual location | Typos, retired locations, wrong warehouse | High |
| C3 | `po_line_number` blank on a PO row | Rows that fall out of every PO-based join | Medium |
| C4 | `heat` blank on a part that requires a heat lot | Traceability gap on heat-controlled material | High |
| C5 | `unit_of_measure` blank or not in the allowed set | Corrupt quantity semantics | Medium |
| C6 | `employee_number = 0` or `user_id` blank | Unattributable rows | Medium |
| C7 | `received_date` in the future or absurdly in the past | Clock or data-entry error | Low |
| C8 | `weight_quantity` zero or negative | Zero-quantity labels | Medium |
| C9 | `qty_ordered` present but `remaining_quantity` stale | Local mirror drift from ERP | Low |
| C10 | Non-PO row missing `vendor_name` | Incomplete non-PO records | Low |

**Note on C4:** the repo already has `lot_markings`, `lot_marking_part_scope`, and `dunnage_no_label_parts`, so "does this part require a heat lot" is answerable locally without touching Visual.

# 4. D. Location Integrity Beyond Drift

The current module only asks "did the location change?". These ask "is the location sane?".

## D1 — Material still in a staging or receiving location after N days

**Catches:** skids that arrived, were staged, and were never put away. This is the most common real-world location problem and it is invisible to a drift-only check because the location never changed.

**Method:** flag rows whose current location is `RECV`, a `V-STAGE`-style location, or any location in a configurable "transient" list, where `received_date` is older than a configurable threshold.

**Effort:** S. **Value:** very high — arguably higher than drift detection.
**Risk:** low.

## D2 — Destination is an ignored or transient location

The ignored-locations setting exists. Extend its use: if transfer evidence points into an ignored location, that currently yields `IgnoredBySettings`. Surface the reason in the exception list instead of counting it as unresolved noise.

**Effort:** S. **Value:** medium.

## D3 — Location in the wrong warehouse or plant

**Catches:** transfer evidence that resolves to a location outside the expected warehouse prefix for the receiving context.

**Effort:** S. **Value:** medium. **Risk:** medium (needs a confirmed warehouse-prefix rule).

## D4 — Current on-hand quantity does not match the labeled quantity at that location

**Catches:** the material is at the right location but in the wrong amount — usually part of the skid was consumed without a transaction.

**Method:** compare `CR_PART_LOCATION` quantity for `(part, location)` against the sum of MTM row quantities placed there.

**Effort:** M. **Value:** high. **Risk:** medium-high — current on-hand includes material from other receipts, so this detects *aggregate* mismatch, not per-skid mismatch. Must be presented as informational, never as a location correction.

## D5 — Ignored-location list hygiene

Report ignored locations that no longer exist in Visual or have had no activity for N days, so the list does not silently suppress real findings.

**Effort:** S. **Value:** low-medium.

## D6 — More destination locations than skids

**Catches:** transfer evidence that spreads one receipt across more locations than there are physical skids, which usually means the movement window pulled in a neighboring receipt.

**Effort:** S once allocation runs. **Value:** medium. Related to the surplus case in `LG-16`.

# 5. E. Traceability, Audit, And Disposition

Every detector above produces findings. Findings need a lifecycle or they become noise nobody reads.

## E1 — Reconciliation run history

Persist each run: user, timestamp, date range, rows scanned, counts per outcome, duration, run id.

**Why:** answers "did anyone run this this week?" and gives `AR-10` something to correlate with.

**Effort:** M (new table + SPs). **Value:** high.

## E2 — Per-row change audit

For every written location change: old location, new location, resolution at write time, evidence (transfer ids, Visual user, transfer time), acting user, timestamp.

**Why:** without this, no one can defend a location change to an auditor, and `LG-12` (idempotency) is impossible.

**Effort:** M. **Value:** very high. **Depends on:** `DS-07`.

## E3 — Undo a run

Reverse the location changes from one run, gated to the original operator or an admin, and only when no later change touched the same row.

**Effort:** M. **Value:** high (it is what makes bulk apply safe to approve).
**Risk:** medium — needs the E2 audit to be trustworthy first.

## E4 — Export exceptions

Export the current finding set to CSV or XLSX with reason, evidence, and proposed action.

**Effort:** S. **Value:** high — most supervisors will triage in Excel before acting.

## E5 — Finding disposition

Per finding: `Open`, `Acknowledged`, `Ignored with reason`, `Resolved`. Persisted so a finding does not reappear every run.

**Why:** today every run rediscovers everything, which trains users to ignore the list.

**Effort:** M. **Value:** very high. **Depends on:** `E1`, `E2`.

## E6 — Assign a finding to a user

Lightweight ownership ("you own this one") for teamwork.

**Effort:** S once E5 exists. **Value:** medium.

# 6. F. Workflow Quality Of Life

## F1 — Date range presets and remembered range

Today / Yesterday / Last 7 days / Last 30 days / This month / Custom. Remember the last used range per user (contrast with `Review.md` 's zero-config behavior — this is deliberate convenience, so confirm it in `UI-02` ).

**Effort:** S. **Value:** high (daily users will run the same range every morning).

## F2 — Exception dashboard card

A count badge per exception family on the landing page ("12 need location review, 3 unlabeled, 1 UOM mismatch"), so users know whether running it is worth their time.

**Effort:** S. **Value:** high.

## F3 — Filter, group, and column chooser

Filter by family, resolution, part, PO, employee, source. Group by part/PO. Show/hide columns like `Module_Reprint` 's column chooser.

**Effort:** S-M. **Value:** medium-high. **Depends on:** `UI-05` , `UI-06` .

## F4 — Bulk select and apply with confirmation

Checkbox grid plus "Apply Selected" with a confirmation dialog listing the exact row count and the distinct destinations.

**Effort:** S (the grid work is already needed for `UI-03` ). **Value:** high.

## F5 — Dry-run indicator

A persistent, unmissable "Preview only — nothing written" vs "Applying" state.

**Effort:** S. **Value:** medium (prevents the worst failure mode: silent bulk writes).

## F6 — Keyboard-first operation

Next/previous finding, apply, skip, re-run, toggle filter. Warehouse users work fast and resent mouse-only screens.

**Effort:** S. **Value:** medium-high.

## F7 — Deep links to the owning surface

From a finding, jump to the row in Receiving Edit Mode, or to `Module_Reprint` for a reprint, or to the Visual transaction in a read-only view.

**Effort:** M. **Value:** high — turns the exception list into an actionable queue instead of a report.

## F8 — "Why is this row here?" explanation panel

Every finding must be explainable in one sentence derived from its evidence, without the user reading code. The current `Details` strings already do this; make it a first-class pane.

**Effort:** S. **Value:** high.

## F9 — Sticky run scope

Show the active date range and filter in the header at all times, so a screenshot of a finding is self-documenting.

**Effort:** S. **Value:** medium.

# 7. G. Automation And Alerting

The repo already has `reporting_scheduled_reports`, so scheduling infrastructure exists.

## G1 — Scheduled nightly scan

Run the read-only detectors nightly over the previous day + unresolved findings, persist results via `E1`, notify on new findings.

**Effort:** M (detectors must be read-only and idempotent first). **Value:** very high.

## G2 — Threshold alerts

Notify (email or Teams) when new exceptions exceed a threshold, for example "more than 5 unlabeled receipts today".

**Effort:** S once G1 exists. **Value:** high.

## G3 — Post-archive trigger

Offer a scan after `sp_Receiving_LabelData_ClearToHistory` completes for a load, so location drift is caught the same day instead of whenever someone remembers to look.

**Effort:** M. **Value:** high. **Risk:** medium — must not block the archive path; run asynchronously and never fail the parent operation.

## G4 — Workstation idle scan

Run a read-only scan when the app has been idle, surfacing a badge. Lower value; listed for completeness.

# 8. H. Analytics And Reporting

## H1 — Reconciliation accuracy trend

Per run: changes proposed vs changes accepted vs changes ignored. Trending acceptance shows whether
the algorithm is trusted.

**Effort:** S once `E1` exists. **Value:** high (this is the metric that justifies the module).

## H2 — Time-to-final-location

Median elapsed time from `received_date` to the transfer that moved material out of the receiving location. A directly useful operational KPI.

**Effort:** M. **Value:** high.

## H3 — Drift by location, employee, or shift

Which locations and which operators generate the most drift or the most unlabeled receipts. Use for training, not punishment — state that in the UI copy.

**Effort:** S once `E1` / `E2` exist. **Value:** medium-high. **Risk:** low, but governance-sensitive.

## H4 — Unresolved finding aging

How long findings stay open. Exposes whether the exception list is actually being worked.

**Effort:** S once `E5` exists. **Value:** high.

## H5 — Reprint volume trend

Reprints per 100 receipts, per employee, per part. A spike usually means a printer, template, or trainer problem.

**Effort:** S once `A2b` exists. **Value:** medium.

# 9. Prioritized Roadmap

Ordering assumes the module extraction happens first with current behavior preserved.

## P0 — Ship with the refactor (no new schema)

| ID | Feature | Why P0 |
|---|---|---|
| C1, C2 | Blank / invalid initial location | Free checks over already-loaded rows |
| C4 | Missing heat lot where required | Uses existing `lot_markings` tables |
| B1 | PO line missing or wrong part | One read-only query, catches real entry errors |
| B5 | UOM mismatch | Protects every quantity comparison in the module |
| D1 | Stuck in staging/receiving | Highest real-world value, no schema change |
| LG-11 support | Reprint-row exclusion with reporting | Correctness prerequisite |
| F2 | Exception dashboard counts | Makes the module worth opening |
| F8 | Explanation panel | Makes findings trustworthy |

| ID | Feature | Why P0 |
|---|---|---|
| E4 | Export | Lets supervisors triage immediately |

## P1 — First follow-up release (needs approval for schema)

| ID | Feature | Dependency |
|---|---|---|
| E1, E2 | Run history + per-row audit | Schema |
| E5 | Finding disposition | E1, E2 |
| A1 | Missing label (needs ERP receipt quantity) | `LG-02` |
| A2b, A3 | Print log + duplicate detection | Schema |
| F3, F4, F6 | Grid ergonomics, bulk apply, keyboard | UI decisions |
| H2 | Time-to-final-location | E1 |

## P2 — Later

| ID | Feature | Note |
|---|---|---|
| E3 | Undo a run | Only after E2 is trusted |
| G1, G2 | Scheduled scan + alerts | Detectors must be read-only first |
| A4, A5, A6 | Split integrity, reprint volume, quality hold | Cheap once print log exists |
| B2, B3, B4 | PO state, over-receipt, date window | Needs ERP receipt semantics |
| D4 | On-hand vs labeled at location | Present as informational only |
| H1, H3, H4, H5 | Analytics | Reporting module surface |
| G3 | Post-archive trigger | Async, non-blocking |
| F7 | Deep links | Touches other modules |

# 10. Data Prerequisites

Grouped by approval cost.

## No schema change

- All of section C.
- B1, B5 (read-only Visual queries only).
- D1, D2, D3, D6 (use existing locations + transfer evidence).
- E4, F1, F2, F5, F6, F8, F9.
- H1/H2/H3 can be computed from logs but are much weaker without `E1`.

## New columns or tables (requires explicit approval)

| Prerequisite | Enables | Why it cannot be avoided |
|---|---|---|
| `receiving_label_print_log` (or `printed_at` + `print_count`) | A2, A5, H5 | There is no print audit today |
| `recon_run` table + SPs | E1, G1, G2, H1, H3 | Run history must outlive a session |
| `recon_change` audit table + SPs | E2, E3, H4 | Location changes need provenance |
| `recon_finding` (disposition) table + SPs | E5, E6, H4 | Otherwise every run rediscovers everything |

## Blocked on business data decisions

| Prerequisite | Enables | Blocked by |
|---|---|---|
| True ERP receipt quantity per PO line | A1, B3 | `LG-02` (receiver data source) |
| Confirmed "transient location" list | D1 | Business input on which locations count as staging |
| Confirmed warehouse-prefix rule | D3 | Business input |

| Prerequisite | Enables | Blocked by |
|---|---|---|
| Confirmed "assy" meaning | `LG-18` | Business input — may add `WIP_ISSUE_DETAIL` as a source |

# 11. Explicitly Out Of Scope

Listed so the spec author does not quietly expand the refactor.

- Writing anything to Infor Visual. This includes any "mark as audited" or "flag" column in ERP.
- Automatic printing or automatic reprinting. The module detects; the user acts.
- Automatic bulk location writes without human review.
- Editing label data (part, quantity, PO) from this module. Edits belong to Receiving Edit Mode.
- Dunnage and Volvo label integrity. Design the seam ( `SC-08` ) but do not implement.
- Replacing `Module_Reprint` . Reprint actions deep-link into it.
- Machine-vision or physical verification of what is on a skid.
- Any change to stored procedure signatures that other screens depend on, unless called out and reviewed individually.